



- Microsoft Visual Studio
- NVIDIA CUDA Toolkit
- NVIDIA cuDNN
- Python (if it is not installed)
- Tensorflow

. . .

## Step 2: Download Visual Studio Express

### Visual Studio is a Prerequisite for CUDA Toolkit

According to NVIDIA, the visual studio must be installed on your System before you can use the NVIDIA CUDA Toolkit(You can use this link to access the prerequisite).



If you attempt to download and install CUDA Toolkit for Windows without having first installed Visual Studio. This error notification will pop up “No supported version of Visual Studio was found. Some components of the CUDA Toolkit will not work properly”. There is Visual Studio Code and Visual Studio( Kindly note the difference)

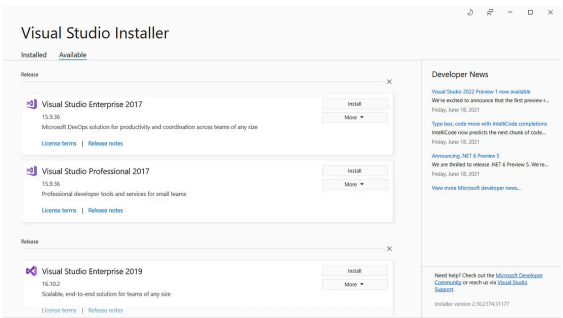
### Selecting and downloading Visual Studio Express

The most recent version of Visual Studio ( which is free) is the Visual Studio Express Community Version 2019. You can access the software with this link.



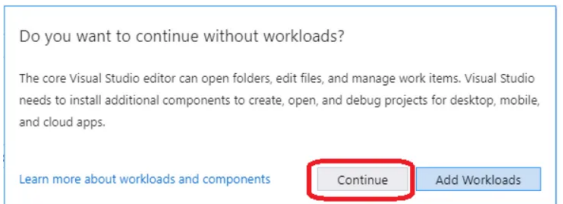
Pictorial Overview of Visual Studio Express (Image by author)

Kindly click on the download community 2019 link, your download will start. Once your download has completed, you can install it like any other software. Once you click on install/open, you will be presented with this image



Installation window for Visual Studio Community 2019. (Image by author)

Kindly click on Available and install Visual Studio Community 2019( I have installed Visual Studio Community 2017). During installation, Visual Studio prompts you to either continue without workloads or add workloads. Kindly click on ‘Continue’



Source

Kindly reboot/restart your PC after installing Visual Studio, this will prevent Visual Studio was still operating and was preventing the installation of CUDA Toolkit error.

## Step 3: Download CUDA Toolkit for Windows 10

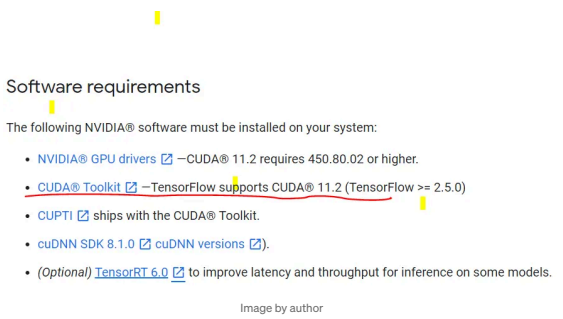


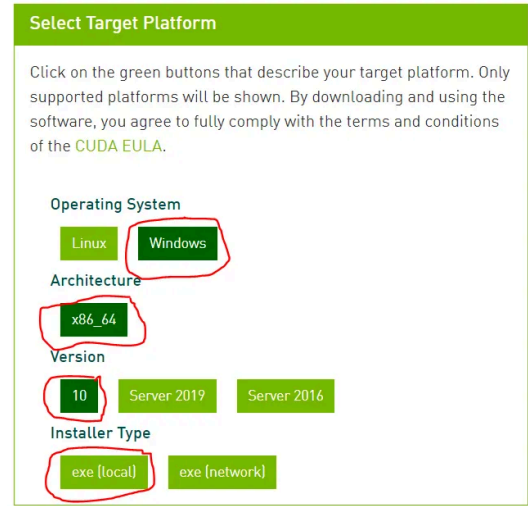
Image by author

Tensorflow ≥2.5.0 requires CUDA Toolkit 11.2 Version, you can access the software with this link

### Archived Releases

CUDA Toolkit 11.3.0 (April 2021), Versioned Online Documentation  
CUDA Toolkit 11.2.2 (March 2021), Versioned Online Documentation  
CUDA Toolkit 11.2.1 (Feb 2021), Versioned Online Documentation  
CUDA Toolkit 11.2.0 (Dec 2020), Versioned Online Documentation  
CUDA Toolkit 11.1.1 (Oct 2020), Versioned Online Documentation

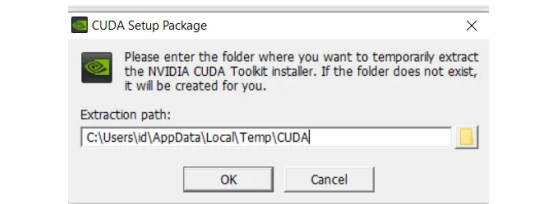
Click on CUDA Toolkit 11.2.0 and download CUDA Toolkit 11.2.0).



Select your target platform, click on the red circled symbols above and click on Download.

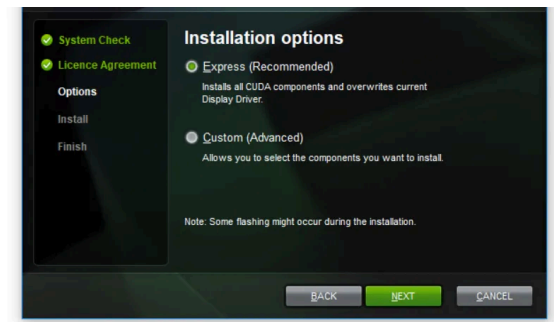


Running the base installer which has been downloaded will produce the CUDA Setup window, as shown below



CUDA Setup Packages for CUDA base installer. Image by author

Click on ok ( don't change the directory), CUDA installer will be extracted to your PC. NVIDIA CUDA Toolkit installation will begin, with a confirmation message. The installer will check if your system meets the requirement, and you will be presented with the screen below.

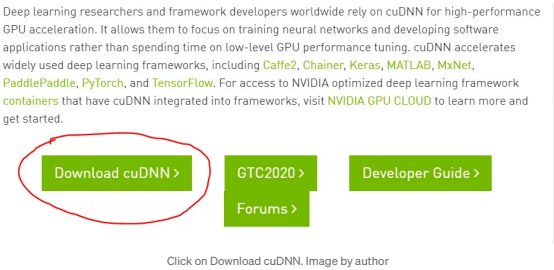


Click on the Express Installation option and click on the Next button. [Source](#)

Just keep clicking on the Next button until you get to the last step( Finish), and click on launch Samples.

#### Step 5: Download and Install cuDNN

Now that CUDA 11.2 has been installed, the next step is to find a compatible version of cuDNN. Tensorflow  $\geq 2.5.0$  requires the cuDNN SDK 8.1.0 version. Use this link to access [NVIDIA cuDNN](#).



Click on Download cuDNN. Image by author

You will need to become a member of the NVIDIA Developer Program to download cuDNN( it is free to join).



cuDNN is a GPU-accelerated set of primitives for deep neural networks.

**Agree To the Terms of the cuDNN Software License Agreement**

Please refer to the [Installation Guide](#) for release prerequisites, including supported GPU architectures and compute capabilities, before downloading. For more information, refer to the cuDNN Developer Guide, Installation Guide and Release Notes on the [Deep Learning SDK Documentation](#) web page.

[Download cuDNN v8.2.1 \(June 7th, 2021\) for CUDA 11.x](#)

[Download cuDNN v8.2.1 \(June 7th, 2021\) for CUDA 10.2](#)

[Archived cuDNN Releases](#)

They are

- cudnn64\_8.dll
- cudnn.h
- cudnn.lib



The screenshot shows a Windows File Explorer window with the address bar displaying 'Search Results in Downloads' and the search query 'cuda\*.dll'. The file list contains the following entries:

| Name                   | Date modified     | Type                  | Size      |
|------------------------|-------------------|-----------------------|-----------|
| cudadv_api_infer64.dll | 6/1/2021 10:05 AM | Application extens... | 172 KB    |
| cudadv_api_train64.dll | 6/1/2021 10:05 AM | Application extens... | 85.0 KB   |
| cudadv_api_infer64.dll | 6/1/2021 10:05 AM | Application extens... | 60.241 KB |
| cudadv_api_train64.dll | 6/1/2021 10:05 AM | Application extens... | 16.444 KB |
| cudadv_api_infer64.dll | 6/1/2021 10:05 AM | Application extens... | 267.44 KB |
| cudadv_api_train64.dll | 6/1/2021 10:05 AM | Application extens... | 57.12 KB  |
| cudadv64.dll           | 6/1/2021 10:05 AM | Application extens... | 219 KB    |

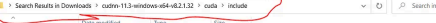
The file 'cudadv64.dll' is highlighted with a red box.

You will need to copy cudnn64\_8.dll directly from the bin sub-folder path. The bin sub-folder is located inside cuda which is a subfolder inside cuDNN.

| Name                  | Date modified      | Type                  | Size      |
|-----------------------|--------------------|-----------------------|-----------|
| ctc                   | 6/20/2021 4:15 PM  | File folder           |           |
| cuda-prov             | 6/20/2021 4:15 PM  | File folder           |           |
| bin2.exe              | 12/1/2020 11:24 AM | Application           | 201 KB    |
| compute-sanitizer.bat | 11/27/2020 2:32 PM | Windows Batch File    | 1 KB      |
| cu-rt.exe             | 12/1/2020 11:24 AM | Application           | 180 KB    |
| cuda64_11.dll         | 12/2/2020 8:25 AM  | Application extens... | 104,852 B |
| cuda64_11.dll         | 12/2/2020 8:25 AM  | Application extens... | 169,096 B |
| cuda64_11.exe         | 12/1/2020 11:24 AM | Application           | 4,576 KB  |
| cuda-memcheck.exe     | 12/2/2020 11:24 AM | Application           | 366 KB    |
| cuda64_11.dll         | 12/2/2020 7:53 AM  | Application extens... | 385 KB    |
| cuda64_11.dll         | 12/2/2020 7:53 AM  | Application extens... | 454 KB    |
| cuda64_11.dll         | 6/21/2021 10:05 AM | Application extens... | 219,109 B |
| cuda64_11.dll         | 12/2/2020 7:53 AM  | Application extens... | 183,478 B |
| cuda64_11.dll         | 12/2/2020 7:53 AM  | Application extens... | 253 KB    |
| cuda64_11.dll         | 12/2/2020 11:24 AM | Application extens... | 1,264 KB  |
| cudaobj.exe           | 12/1/2020 11:24 AM | Application           | 2,811 KB  |

44 items

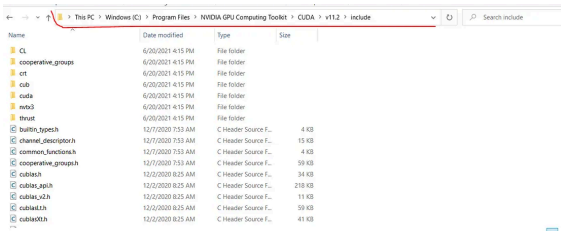
## 5.2 cudnn.h



The screenshot shows the Splunk Search interface. The search bar contains the query `cassandra* | headlimit 10000`. The search results table is as follows:

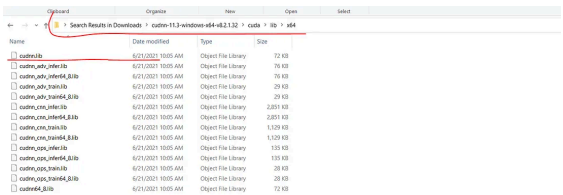
| Name                   | Time modified      | Type              | Size  |
|------------------------|--------------------|-------------------|-------|
| cassandra              | 6/21/2021 10:05 AM | C:Header Schema F | 3 KB  |
| cassandra_auth_index   | 6/21/2021 10:05 AM | C:Header Schema F | 2 KB  |
| cassandra_auth_streams | 6/21/2021 10:05 AM | C:Header Schema F | 19 KB |
| cassandra_backend      | 6/21/2021 10:05 AM | C:Header Schema F | 17 KB |
| cassandra_index        | 6/21/2021 10:05 AM | C:Header Schema F | 29 KB |
| cassandra_logs_streams | 6/21/2021 10:05 AM | C:Header Schema F | 19 KB |
| cassandra_logs_streams | 6/21/2021 10:05 AM | C:Header Schema F | 50 KB |
| cassandra_streams      | 6/21/2021 10:05 AM | C:Header Schema F | 20 KB |
| cassandra_streams      | 6/21/2021 10:05 AM | C:Header Schema F | 18 KB |

You will need to copy cudnn.h into include subfolder located inside CUDA. The CUDA folder is located NVIDIA GPU Computing Toolkit Folder.



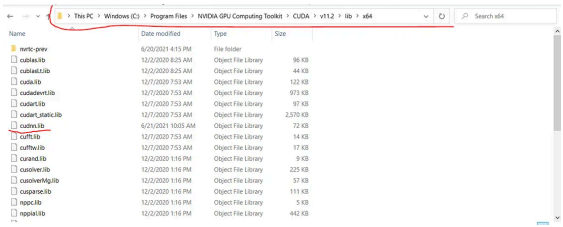
### 5.3 cudnn.lib

You will need to copy cudnn.lib from the x64 subfolder which is located inside lib subfolder. The lib subfolder is located inside cuDNN.



Copy cudnn.lib. Image by author

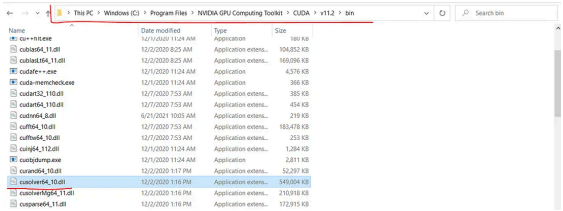




Paste the copied cudnn.lib into the lib subfolder located inside CUDA. Image by author

### Renaming cursolver64\_11.dll

You will need to rename cusolver64\_11.dll to cusolver64\_10.dll, else you will get this error *“Could not load dynamic library ‘cusolver64\_10.dll’; derror: cusolver64\_10.dll not found*’ when you try to leverage your GPU for Tensorflow training.



I have renamed cursolver64\_11.dll to cursolver64\_10.dll (you need to do this). Image by author

### Step 5.3: Confirming that CUDA environment variables are set in Windows

For CUDA to function properly, you will need to ensure that CUDA environment variables are set in your PC's Path. To confirm, you will need to check the Environment Variables. This can be found by following these steps

Control Panel -> System and Security -> System-> Advanced System settings.

This will open up a window, click on the Advanced tab, as shown below

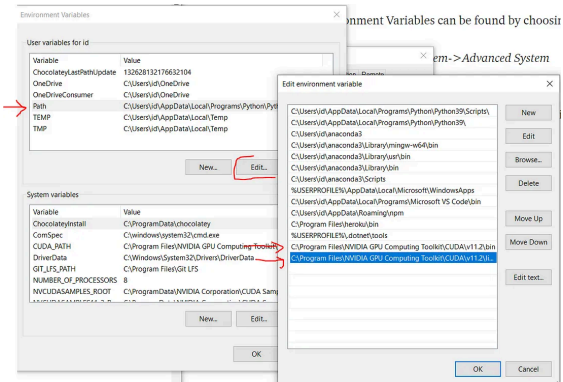
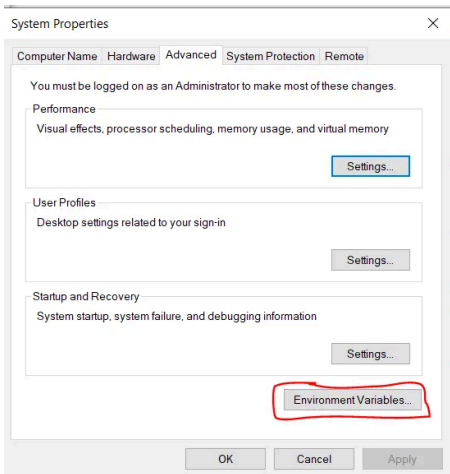


Image by author

These paths are automatically added, but in case you can't find them. Just type in C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v11.2\bin and C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v11.2\libnvvp.

### Step 6: Install Python (In case you haven't done that before)

I recommend using Anaconda to download python, you can access anaconda with this [link](#).

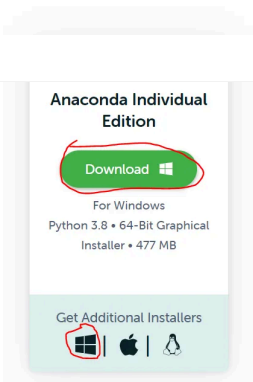
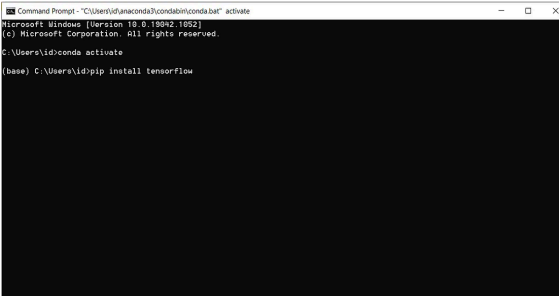


Image by author

It is pretty easy to install anaconda, instructions to easily install will be displayed once you click on install.

### Step 7: Installing Tensorflow (If it is not installed)



Open your terminal, activate conda and pip install TensorFlow. Image by author

#### Step 8: Test Installation of TensorFlow and its access to GPU

Open your terminal( command prompt), type conda activate and type python ( to enter python interactive mode)

```
# importing tensorflow package
```

```
import tensorflow as tf
```

Run this code to test CUDA support for your Tensorflow installation,

```
tf.test.is_built_with_cuda()
```

To confirm that the GPU on the system is accessible by Tensorflow, you can test with this code

```
tf.test.is_gpu_available(cuda_only=False,min_cuda_compute_capability=None
```

It will return *True* after few minutes.

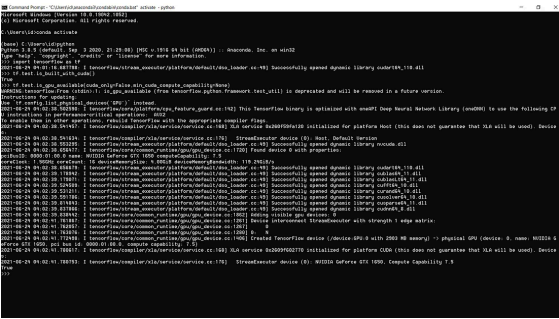


Image by author

#### Optional Step

You will likely encounter this error “Not creating XLA devices, tf\_xla\_enable\_xla\_devices not set”. XLA helps to speed up training on GPU, to set tf\_xla\_enable\_xla\_devices you can follow this step

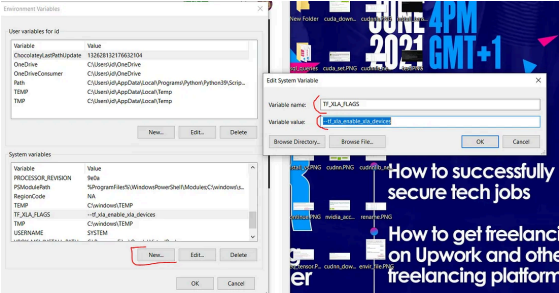


Image by author

Navigate to Environment Variables, navigate to System Variables and click on the New button. Fill in the Variable name and Variable value as shown above.

#### Conclusions

This article aimed to build on this [old article](#) by Dr Joanne Kitson, and provide an updated instruction for installing TensorFlow with NVIDIA CUDA, cuDNN and GPU support on windows without errors. I might write on how to Install CUDA,cuDNN and GPU on Linux Ubuntu Distro in my next article. Thanks for reading up to this point, kindly subscribe to my newsletter for early access to my articles.

#### References

- <https://towardsdatascience.com/installing-tensorflow-with-cuda-cudnn-and-gpu-support-on-windows-10-60693e46e781> ( Inspired the write-up and served as a guide while installing TensorFlow)
- [Python – Tensorflow GPU Could not load dynamic library 'cusolver64\\_10.dll'; dlerror: cusolver64\\_10.dll not found – Stack Overflow](#)
- [cuda install](#)

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HackerCop

Sep 21, 2021

...

You no longer need to rename cusolver64\_11.dll to cusolver64\_10.dll this is outdated. if you do you will get an error saying it cannot find cusolver64\_11.dll If you could remove that part it would be great.

👤 5

Reply



Ramganeshlankada

Dec 18, 2022 (edited)

...

you've given a very detailed instructions sir. thank you soo much. keep pushing these kind of articles

👤

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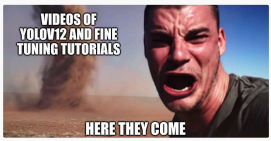


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